PADHAKU
Exploration · Class 9 Science



CHAPTER 12

Patterns in Life: Diversity and Classification

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🤔 Think It Over

1 **What do you understand by biodiversity?**

✔ ANSWER

Biodiversity is the huge variety of living organisms on the Earth — from microscopic organisms to giant trees, living in many different habitats (snow-clad mountains to coral reefs).

2 **How does the grouping of organisms help us understand diversity?**

✔ ANSWER

Grouping puts organisms with similar features together. It makes the huge variety easy to study, shows similarities and differences, and tells us how organisms are related to one another.

3 **On what basis, are plants and animals classified?**

✔ ANSWER

On the basis of shared features, such as:

- external features (shape, size, body organisation)
- mode of nutrition (autotrophic/heterotrophic)
- internal structures and tissues

- cell structure (unicellular/multicellular, prokaryote/eukaryote, cell wall)
- ecological role, reproduction and genetic (DNA) similarity

4 **How does classification help address problems in farming?**

✔ ANSWER

Classification helps farmers identify crop varieties with useful features (drought tolerance, pest resistance, growing in poor soil), and tell useful insects from pests. Growing diverse varieties reduces the risk of crop failure and strengthens food security.

🧪 Activity 12.1 · 12.3 How to Classify Organisms?

1 **Which animals can you identify in the picture (Fig. 12.2)? Where are they seen?**

💡 SAMPLE ANSWER

Day side: peacock, spotted deer, leopard, bear, tiger, lion-tailed macaque, langur (monkey), eagle, rabbit, small birds, butterfly, dragonfly.

Night side: owl, bat, snake (python), porcupine, slender loris, civet, frog, crocodile, fireflies.

Where: on the forest floor (deer, tiger, bear, porcupine), on trees (monkeys, owl, snake, loris), in the air (eagle, bat, birds, insects), in/near water (crocodile, frog). (check the picture in your book)

1 (iii) Which animals seem active (i) during the day? (ii) during the night? (iii) both during the day and the night?

💡 SAMPLE ANSWER

(i) Day: peacock, deer, tiger, monkeys, eagle, rabbit, butterflies.

(ii) Night: owl, bat, porcupine, slender loris, civet, fireflies.

(iii) Both: animals like the snake, frog, crocodile and leopard can be active by day and by night. (check the picture in your book)

2 Record your observations in Table 12.1.

💡 SAMPLE ANSWER

Organism	Where seen	Active when	Visible feature
Owl	Tree	Night	Feathers
Tiger	Forest floor	Day	Striped fur
Bat	Flying in air	Night	Wings of skin
Peacock	Ground	Day	Colourful feathers

Crocodile	Water	Day/night	Scaly skin
Butterfly	Flying near ground	Day	Wings, six legs

3 **Group the same organisms in more than one way, changing the criterion each time (Table 12.2).**

💡 **SAMPLE ANSWER**

Grouping criterion	Organisms	Feature that helped
Carnivore	Eagle, tiger, leopard	Eating habits
Herbivore	Deer, rabbit	Eating habits
Nocturnal	Owl, bat, loris, porcupine	Time of activity
Can fly	Eagle, owl, bat, butterfly	Wings
Live in/near water	Crocodile, frog	Habitat

The same organism (e.g. owl) can fit into different groups depending on the criterion.

⭐ Pause and Ponder · 12.3.1 Some criteria to classify living organisms

1 **If many organisms share common features, could they also share a common ancestry?**

✔ **ANSWER**

Yes. Similar features suggest that the organisms have evolved from a common ancestor. The more features they share, the more closely they are likely to be related.

Activity 12.2 · 12.4 The Need for Classification

? How are species distributed within a forest? Which plants and animals are closely linked?

✔ ANSWER

Species are found in different parts of the forest depending on their needs, e.g. different hornbill species live where suitable tree size and fruits are available. Hornbills are closely linked to large, old trees (nesting cavities) and fruit trees (food).

? How does classifying the four hornbill species help us understand biodiversity?

💡 SAMPLE ANSWER

It shows how closely related species differ in their needs (nest trees, fruits) and where each lives. This helps scientists track each species and plan its conservation.

(i) How can scientists keep track of so many species?

✔ ANSWER

By classifying them — grouping species by shared features, giving each a scientific name, and recording where and how each lives.

(ii) **The four hornbills look similar in some ways. What features can help scientists distinguish them from one another?**

💡 SAMPLE ANSWER

Size, colour of the feathers, shape and colour of the beak and casque (the helmet on the beak), colour of the neck and throat, calls, and the fruits and trees they use.

(iii) **What would happen if the large, old trees disappeared from the forest?**

✔ ANSWER

Hornbills nest only in large, old trees with cavities, so they could not breed and their numbers would fall. Seed dispersal by hornbills would also drop, affecting the forest and other species linked to them.

🧪 Activity 12.3 · 12.6 Five Kingdom Classification

2 **Study the concept map (Fig. 12.5). List the criteria which form the basis of five kingdom classification.**

✔ ANSWER

- Cell type: prokaryote or eukaryote
- Level of organisation: unicellular or multicellular
- Cell structure: cell wall present or absent (and made of chitin or cellulose)
- Mode of nutrition: autotrophic or heterotrophic
- Ecological role: producer, consumer or decomposer

3 **Compare the criteria you listed with: cell type, cell structure, level of organisation, mode of nutrition.**

✔ ANSWER

They match. The concept map also shows the ecological role (producers, consumers, decomposers) and the material of the cell wall (chitin in fungi, cellulose in plants).

🧪 Activity 12.4 · 12.6.1 Kingdom Monera

2 **Observe slides of bacteria and cyanobacteria and compare with Fig. 12.6. What do you observe?**

✔ ANSWER

Observation: very small single cells (rod, round or spiral bacteria; cyanobacteria often in chains/threads, bluish-green). No distinct nucleus is seen.

Conclusion: they are unicellular prokaryotes and belong to the kingdom Monera.

🧪 Activity 12.5 · 12.6.2 Kingdom Protista

7 **Do you notice moving organisms in the drop of hay infusion under the microscope? Can you identify them by comparing them with Fig. 12.7?**

✔ ANSWER

Observation: yes, many tiny organisms move about — slipper-shaped Paramecium (moves with cilia), Amoeba (changes shape, uses false feet) and green Euglena (moves with a flagellum).

Conclusion: these are unicellular eukaryotes (protists) that live in water.

⭐ Pause and Ponder · 12.6.2 Kingdom Protista

2 **How can a single-celled organism carry out all its life processes when billions of cells are required to perform similar functions in multicellular organisms like us?**

✔ ANSWER

In a single-celled organism, the one cell does everything — taking food, digestion, respiration, removing waste, movement and reproduction. Its organelles (e.g. food vacuole, contractile vacuole, cilia) do the jobs that tissues and organs do in us. Being small, it exchanges materials directly with its surroundings.

⭐ Bridging Science and Society · Thallophyta (lichens)

? **Have you seen white-green patches on tree trunks, damp walls or stones? What do you think about these patches?**

✔ ANSWER

These patches are lichens — an alga and a fungus living together. The alga makes food, the fungus gives protection. They grow only in clean air, so they indicate a pollution-free environment (bioindicators).

🧪 Activity 12.6 · Bryophyta

2 **Observe bryophytes under a dissecting microscope. How do they differ from the leaves that you usually observe in plants around you?**

✔ ANSWER

Observation: the 'leaves' are very small, thin and simple, with no stalk and no veins. The plant is fixed by thread-like rhizoids, not true roots.

Conclusion: bryophytes have only leaf-like and stem-like structures, without vascular tissue — not true leaves.

? In-text questions · Pteridophyta

? Look at Fig. 12.11a (fern). What changes do you observe in the plant structure as compared to thallophytes and bryophytes?

✔ ANSWER

A fern has true roots, stem and leaves; the leaves are large and divided. Thallophytes have only a thallus and bryophytes only rhizoids and simple leaf-like structures.

? How do you think transport of water and food takes place in all parts of these plants? How can you test this explanation?

✔ ANSWER

Ferns have special transport (vascular) tissues. Test: cut a thin cross section of a fern stem (or use a permanent slide) and see it under a microscope — xylem (water) and phloem (food) are seen.

⭐ Pause and Ponder · Pteridophyta

3 **Which plant features reduce their dependence on water but still require moist conditions?**

✔ ANSWER

Rhizoids, stem- and leaf-like structures (bryophytes), and true roots, stems, leaves with vascular tissue (pteridophytes) reduce dependence on water. But both still need water/moisture for reproduction, as male cells must swim to the female cells.

4 **Why do taller plants need specialised transport tissues?**

✔ ANSWER

In tall plants water and minerals must reach leaves far above the roots, and food must go from leaves to all parts. Simple diffusion is too slow over long distances, so xylem and phloem are needed. They also give support.

5 **How do seeds and fruits affect where and how plants can survive?**

✔ ANSWER

Seeds protect the embryo and store food, so plants can survive dry or cold periods and do not need water for reproduction. Fruits help spread seeds to new places. So seed plants (gymnosperms, angiosperms) can live in many more habitats, even dry ones.

🧪 Activity 12.7 · Let us compare

3 **Compare the cross section of a fern stem (Fig. 12.11b) with that of a sunflower stem (Fig. 3.7). What difference do you observe in the vascular tissue?**

💡 SAMPLE ANSWER

- **Fern:** xylem and phloem are present, the xylem lies in the centre surrounded by phloem; the vascular tissue is simple.
- **Sunflower (higher plant):** many vascular bundles arranged in a ring; in each bundle xylem is towards the inside and phloem towards the outside.

Conclusion: both have vascular tissue, but it is more developed and organised in higher (flowering) plants. (check the figures in your book)

🧪 Activity 12.8 · Angiosperms

– **Collect leaves, observe their shape and venation, group them as monocots or dicots,**

and discuss how their structures help them adapt.

💡 SAMPLE ANSWER

- **Parallel venation → monocots:** grass, wheat, maize, banana, bamboo. Long narrow leaves suit open, sunny places and wind.
- **Reticulate (net) venation → dicots:** mango, peepal, hibiscus, neem, rose. Broad leaves catch more sunlight; the vein network gives strong support and transport.
- Thick or waxy leaves reduce water loss in dry places.

🧪 Activity 12.9 · Table 12.3 (fill the blanks)

Thallophyta **Add an advantage / challenge for Thallophyta.**

💡 SAMPLE ANSWER

Challenge: no vascular tissue or true roots, so they cannot grow tall; dry up quickly out of water.

Bryophyta **Add an advantage / challenge for Bryophyta.**

💡 SAMPLE ANSWER

Advantage: rhizoids fix them to the soil; they can grow on rocks and walls.

Challenge: no vascular tissue, so they stay small; need water for reproduction.

Pteridophyta **Add an advantage / challenge for Pteridophyta.**

💡 SAMPLE ANSWER

Advantage: true roots, stem and leaves; can grow taller.

Challenge: no seeds; need moist places to reproduce.

Gymnosperm **Add an advantage / challenge for Gymnosperm.**

💡 SAMPLE ANSWER

Advantage: survive in cold and dry regions; seeds store food for the embryo.

Challenge: seeds are naked (not protected by fruits); slow growth; no flowers to attract pollinators, pollen spread by wind.

Angiosperm **Add an advantage / challenge for Angiosperm.**

💡 SAMPLE ANSWER

Advantage: fruits help seed dispersal to new places; live in a wide range of habitats.

Challenge: depend on pollinators; if pollinators decline, reproduction is affected.

In-text questions · 12.6.5 Kingdom Animalia (invertebrates)

? Sponges: do you think they would be able to survive on land? Why or why not?

✔ ANSWER

No. Sponges get food and oxygen from water flowing through their pores, and water carries away their wastes. On land there is no water flow, so they would dry up and die.

? Cnidarians have a single opening for food intake and waste removal. Can you hypothesise what limitations this body structure might have?

✔ ANSWER

Food cannot be taken in while waste is being removed, so eating and digestion are slow; fresh food and waste may mix. This limits body size and activity.

? Annelids: what advantages might segmentation offer to an organism?

✔ ANSWER

Segments can move one after another, giving flexible, precise movement (the earthworm stretches and shortens part by part). Damage to one segment does not affect the whole body.

? **Arthropods: what advantages can the outer covering (exoskeleton) provide?**

✔ ANSWER

It gives protection, reduces water loss and supports strong muscles, so arthropods can survive in dry and exposed places.

⭐ **Pause and Ponder · Arthropoda**

6 **An earthworm and a beetle both have segmented bodies but the beetle has a hard external skeleton. How does it help the beetle survive?**

✔ ANSWER

The hard exoskeleton protects the beetle from injury and enemies, reduces water loss so it can live in dry places, and supports powerful muscles for fast movement. The earthworm, without it, must stay in moist soil.

12.7.1 The hierarchical nature of classification

? **Observe and analyse Fig. 12.17. What do you infer?**

✔ ANSWER

Classification goes step by step from a large, broad group (kingdom) to small, specific groups down to the species. At each lower level there are fewer organisms but they share more features. E.g. tiger: Animalia → Chordata → Vertebrata → Mammalia → Carnivora → Felidae → Panthera → P. tigris.

Pause and Ponder · 12.9–12.10

7 **Does the term 'biodiversity' relate only to the variety of organisms, or does it encompass other elements?**

✔ ANSWER

It includes more: the variety within a species (e.g. many crop varieties), the variety of species, and the variety of habitats/ecosystems (forests, deserts, coral reefs) along with the interactions among them.

8 **If you find a new organism in a pond, what features will you observe to classify it and why?**

✔ ANSWER

- Cell type (true nucleus or not) → Monera vs others
- Unicellular or multicellular
- Cell wall present or not, and of what
- Mode of nutrition (green/photosynthetic or not)
- Body organisation, movement, and for animals notochord/backbone

These are the basic criteria of the five kingdom system, from broad to detailed.

9 **Why do genetic studies provide deep information about living beings?**

✔ ANSWER

DNA carries the instructions for growth and function of every cell. Comparing DNA shows exactly how similar organisms are and whether they share a common ancestry, even when their outer looks are misleading.

10 **How can changes in climate affect the biodiversity?**

✔ ANSWER

Changes in temperature and rainfall alter habitats. Species that cannot adapt or move may decline or become extinct, and other species that depend on them are also affected, reducing biodiversity.

⭐ Bridging Science and Society · Forests and disasters; Sangai deer

? How do forests with rich and highly diverse flora and fauna work as a barrier and reduce the impact of disasters?

✔ ANSWER

- Mangrove forests reduce damage from cyclones (Odisha super cyclone, 1999).
- Rich biodiversity lowers the risk of tick-borne diseases like Monkey Fever (KFD).
- Soil microorganisms and mangrove soils break down or trap pollutants, protecting water.

? Provide your suggestions to save the phumdis habitat and the Sangai deer.

💡 SAMPLE ANSWER

- Stop pollution and over-use of Loktak lake water; control dams/water levels that harm phumdis.
- Stop hunting and disturbance; protect the national park strictly.

- Involve local people in conservation and create awareness.
- Breed Sangai in protected areas and monitor their numbers.

⭐ Revise, Reflect, Refine

1 **Meena and Hari observed an animal in their garden. Hari called it an insect while Meena said it was an earthworm. Choose the option which confirms that it is an insect.**

(i) Bilateral symmetrical body

(ii) Body with jointed legs

(iii) Cylindrical body

(iv) Body with little segmentation

✔ ANSWER

(ii) Body with jointed legs — jointed appendages are found in arthropods (insects); earthworms have no legs.

2 **Sponges lack true tissues and organs. Which feature of sponge cells supports their classification under the animal kingdom?**

(i) Absence of mitochondria

(ii) Ability to photosynthesise

(iii) Presence of a cell membrane

(iv) Presence of a cell wall

✔ ANSWER

(iii) Presence of a cell membrane — animal cells have only a cell membrane and no cell wall; sponges also do not photosynthesise.

3 **Observe two different animals in your immediate environment. What features help you distinguish between them? How do these features help place them into different groups?**

💡 **SAMPLE ANSWER**

E.g. a **sparrow** and an **ant**.

- Sparrow: backbone, feathers, two legs, wings, beak → vertebrate, class birds.
- Ant: no backbone, six jointed legs, hard exoskeleton, segmented body → invertebrate, Arthropoda (insect).

Presence of backbone, body covering and type of limbs place them in different groups.

4 **How would a scientist justify choosing cellular organisation as a more fundamental basis of classification than the presence of xylem and phloem?**

✔ **ANSWER**

Every organism is made of cells, so cellular organisation (prokaryote/eukaryote, unicellular/multicellular) can be used for all living beings. Xylem and phloem occur only in some plants, so they are useful only for dividing plants, not for all life.

5 **You find an unlabelled slide of a single-celled organism with a well-defined nucleus and multiple cilia. Which group would it most likely belong to? Give reasons.**

✔ ANSWER

Kingdom Protista (likely Paramecium).

- Single-celled → not fungi, plants or animals (multicellular).
- Well-defined (membrane-bound) nucleus → eukaryote, so not Monera.
- Cilia for movement are typical of protists like Paramecium.

6 **How does the diversity of organisms contribute to the balance and stability of an ecosystem?**

✔ ANSWER

Each organism has a role: plants and algae produce food and oxygen, animals pollinate flowers and disperse seeds, fungi and bacteria decompose and recycle nutrients. Many species make many food links, so if one species is lost others can take its place and the ecosystem stays stable.

7 **If all unicellular organisms were grouped into a single kingdom, what problems would arise?**

✔ ANSWER

Very different organisms would be put together: prokaryotes (bacteria, no true nucleus) with eukaryotes (Amoeba, Euglena), autotrophs with heterotrophs, and yeast (a fungus with chitin wall) with protists. The group would not show true relationships or common ancestry.

8 **Why are viruses not placed in any of the five kingdoms? Give reasons.**

✔ ANSWER

- Viruses are not made of cells (acellular); the five kingdoms are based on cell features.
- They have no nucleus, cytoplasm or cell wall; just genetic material in a protein coat.

- They show life only inside a host cell and are inactive outside it.

9 **If asked to revise the five kingdom classification, would you create a separate category for viruses or keep them outside the system? Justify and explain what this shows about scientific classification.**

💡 SAMPLE ANSWER

Keep them **outside** the five kingdoms (in a separate group of acellular, non-cellular entities), because the kingdoms are based on cell features and viruses have no cells and can multiply only inside host cells.

This shows that classification keeps changing as new facts are found — it is an ongoing process (two kingdom → five kingdom → three domain).

10 **Viruses contain genetic material but lack cellular organisation. Which features prevent them from fitting into the five kingdom system? What does this tell us about the limitations of classification systems?**

✔ ANSWER

They lack cells, cell type (prokaryote/eukaryote), cell wall, level of organisation and their own mode of nutrition — the very criteria of the five kingdoms. So any classification system is limited by the criteria it uses; it cannot fit everything and must be revised as knowledge grows.

11 **Both pteridophytes and bryophytes lack flowers and seeds, yet are placed in different groups. Explain using their key features.**

✔ ANSWER

Bryophytes	Pteridophytes
No true roots, stem, leaves; only rhizoids	True roots, stem and leaves
No vascular tissue	Vascular tissue (xylem, phloem) present
Small, grow in moist shady places	Can grow larger

Presence of true organs and vascular tissue places pteridophytes in a separate, more advanced group.

12 **In the classification hierarchy, which group — class or genus — has fewer members but more features in common? Explain.**

✔ ANSWER

Genus. Going down the hierarchy (kingdom → ... → class → order → family → genus → species), each group is smaller and its members share more common features. Genus is below class, so it has fewer members with more similarities.

13 **A new organism shows locomotion and autotrophic nutrition. Which character(s) would help identify it as belonging to Protista?**

✔ ANSWER

It should be **unicellular** and **eukaryotic** (with a membrane-bound nucleus). A single-celled eukaryote that moves and photosynthesises (like Euglena) belongs to Protista. (Plants are autotrophic but do not move and are multicellular; Monera have no true nucleus.)

14 **A researcher identified a unicellular eukaryotic organism as fungi. What identification key would you suggest to keep a unicellular organism in the Kingdom Fungi?**

✔ ANSWER

- Eukaryotic cell with a **cell wall made of chitin**
- No chlorophyll; **heterotrophic**, absorbs nutrients from dead/decaying matter (saprophytic)
- Reproduces by spores/budding

Example: yeast — unicellular, but kept in fungi because of its chitin cell wall.

15 (i) **Organisms and key observations:**

Organism	Key observations
P	Microscopic; no true nucleus; rigid cell covering; survives high salinity and temperature
Q	Multicellular; filamentous body; cell wall present; no chlorophyll; grows on dead organic matter
R	Unicellular; true nucleus; contractile vacuole; moves using flagella; photosynthesis in light but heterotrophic in dark
S	Multicellular; well-differentiated tissues; backbone present; aquatic respiration in early life
T	Acellular; contains genetic material; inactive outside a host cell

(i) Identify one organism that clearly belongs to the Kingdom Fungi. State one observation that supports your answer.

✔ ANSWER

Q. It is multicellular, filamentous, has a cell wall but no chlorophyll, and grows on dead organic matter (saprophyte).

15 (ii) **Which organism would be placed in the Kingdom Monera? Mention one**

characteristic that justifies this placement.

✔ ANSWER

P. It has no true nucleus (prokaryote) and is microscopic. (Its survival in high salt and heat suggests archaea.)

15 (iii) **Organisms R and Q are both eukaryotic, yet they are placed in different kingdoms. Analyse the criteria that separate them.**

✔ ANSWER

- Level of organisation: R unicellular, Q multicellular.
- Nutrition: R can photosynthesise (autotrophic in light); Q has no chlorophyll and absorbs food from dead matter.
- Movement: R moves with flagella; Q does not.

So R → Protista, Q → Fungi.

15 (iv) **Explain why organism S cannot be classified using the mode of nutrition alone.**

✔ ANSWER

S is heterotrophic, but so are fungi, many protists, bacteria and all animals. Nutrition alone cannot separate them. Other features — multicellular, tissues, no cell wall, backbone — are needed to place S in Animalia (vertebrates, likely an amphibian such as a frog).

15 (v) **Organism T does not fit into any of the five kingdoms. Which fundamental characteristic does it lack and what does this reveal about the limitations of classification systems?**

✔ ANSWER

T lacks **cellular organisation** (it is acellular, like a virus). The five kingdom system is based on cell features, so it cannot include such entities. Classification systems are human-made and limited by their criteria; they must change as new knowledge comes.

15 (vi) **If classification were based only on habitat, which organisms might be incorrectly grouped together? Explain the scientific consequences.**

✔ ANSWER

The pond organisms R (a protist) and S (a vertebrate) would be grouped together, and Q with soil bacteria, though they are very different. Unrelated organisms would be put together and related ones separated, so the system would not show true relationships or evolution and would be of little use for study.

15 (vii) **A new organism is multicellular, eukaryotic, lacks chlorophyll and absorbs nutrients from a host externally. Should it be placed under fungi or animalia? Justify.**

✔ ANSWER

Fungi. Fungi absorb digested food from outside the body (absorptive nutrition), like parasitic fungi; animals take food inside (ingest) and digest it. Lack of chlorophyll rules out plants. A cell wall of chitin would confirm it as a fungus.

⭐ The Journey Beyond

1 **Visit a nearby park or water body. Identify ten organisms and classify them into kingdoms.**

💡 SAMPLE ANSWER

Organism	Kingdom
Bacteria in pond water	Monera
Paramecium, Euglena	Protista

Mushroom, bread mould	Fungi
Spirogyra, moss, fern, lotus, grass, neem	Plantae
Fish, frog, duck, snail	Animalia

2 **Interview a gardener or forest worker on how they identify species. Compare with scientific methods.**

💡 **SAMPLE ANSWER**

Gardeners use leaf shape, smell, flowers, bark, local names and uses. Scientists use detailed features, cell structure, DNA and scientific (binomial) names. Both observe features carefully; the scientific method is universal and more exact.

3 **If you were studying the Pakke Tiger Reserve or a bird sanctuary, how would you organise information about the bird species?**

💡 **SAMPLE ANSWER**

Make a table/database for each species: common and scientific name, family, size and colour, food, nesting trees, part of the forest, season seen, and numbers. Group species by family and by feeding habits.

4 **Conduct a survey near a village to identify edible and poisonous mushrooms with the help of community resource persons.**

💡 **SAMPLE ANSWER**

Record local and scientific names, colour, shape, where it grows, and whether it is edible, medicinal or poisonous. Never taste a wild mushroom yourself.

5 **Collect postal stamps on flora and fauna, classify them by your own criteria and organise a philately exhibition.**

💡 **SAMPLE ANSWER**

Possible criteria: plants/animals; kingdom or class (birds, mammals, reptiles, flowers); habitat; endangered or not; Indian or foreign.

6 **Investigate an extinct species and analyse how its disappearance affected dependent organisms and the ecosystem.**

💡 **SAMPLE ANSWER**

Example: the dodo (Mauritius). The seeds of some trees were believed to germinate better after passing through its gut, and their regeneration may have been affected. Losing one species can break food chains and affect other species.

7 **Study the diversity of farm animals used in different types of farming in various parts of the country.**

💡 **SAMPLE ANSWER**

Examples: cows and buffaloes (dairy), bullocks and camels (draught, Rajasthan), yaks (Himalayas), goats and sheep (wool, meat), poultry, fish (aquaculture), honey bees (apiculture).

Quest **What knowledge do you think scientists would come up with further and how would it affect the classification system?**

💡 **SAMPLE ANSWER**

Better DNA studies and new discoveries (new microbes, deep-sea life) will show relationships more clearly. Groups may be split, merged or renamed, and new categories may be added — classification will keep changing.

🎉 **Chapter 12 done! Revise the answers once more tomorrow.**